Assignment 1

Deadline: September 25, 2026

Q1. Compute the empirical and generator-based estimators of one-year transition matrix for the following data.

(a) There are four credit classes, A, B, C, D. At the beginning, there are 20 firms in A, B, C, each.

(b) In one month, two B rated firms migrate to C, and two B rated firms migrate to A , and these firms stay there for the rest of the year.

(c) In three months, three A rated firm migrates to B, and stay there for the rest of the year.

(d) In seven months, two C rated firms migrate to B and stay there for the rest of the year. Furthermore, one C rate firm defaults.

(e) Ten months later, two C rated firms default.

Q2. A company has issued 3 - and 5 -year bonds with a coupon payment of 4.1% per year payable annually. The yields on the bonds (in terms of continuous compounding) are 4.6% and 4.75%, respectively. Assume the risk-free rate is 2.5% for all maturities, and the recovery value is 40% of the par value. Also assume default can only take place halfway through each year, i.e. at year $0.5, 1.5, 2.5$, etc. Compute P_1 and P_2, which are the implied default intensities per year for years 1 to 3 and for years 4 and 5, respectively.

Q3. Compute (1) and (2) in Lecture 2 under the Black-Scholes-Merton geometric Brownian motion model.

A Itô Lemma

In its simplest form, Itô lemma states the following: for an Itô drift-diffusion process

$$dX_t = \mu_t dt + \sigma_t dW_t,$$

and any twice differentiable scalar function $f(t, x)$ of two real variables t and x, one has

$$df\left(t, X_t\right) = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{\sigma_t^2}{2}\frac{\partial^2 f}{\partial x^2}\right) dt + \sigma_t \frac{\partial f}{\partial x} dW_t. \tag{A.1}$$

B Density of Log-normal Process $V(\cdot)$

Consider the following stochastic differential equation

$$\begin{cases} dV_t = rV_t dt + \sigma V_t dW_t, \\ V_0 = 1, \end{cases} \tag{B.1}$$

where r and σ are constant.

Then

$$\ln V_T \ \sim \ N\left(\left(r - \frac{1}{2}\sigma^2\right)T, \sigma^2 T\right), \tag{B.2}$$